

STATS ANALYSIS

An NBA exploratory career statistics analysis

Avg. Reading time: 5 minutes

DATASET

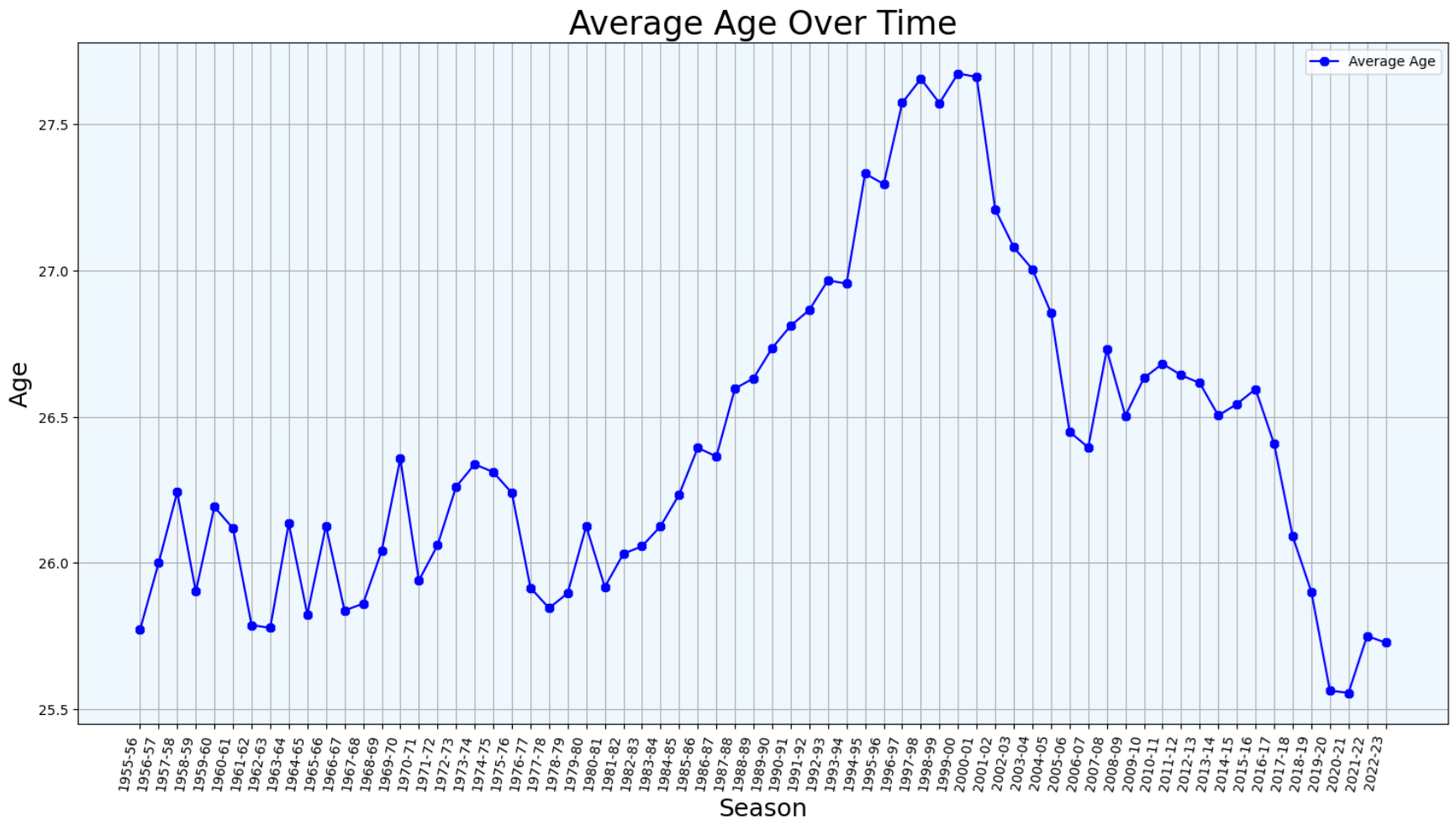
- Data from basketball-reference website
- Use of players per-game and season-total data for every season from 1955-56 to 2022-23
- Total of 27,845 records
- Use of MVP & Championship data
- Implemented in Python on a Jupyter notebook

DATA PREVIEW

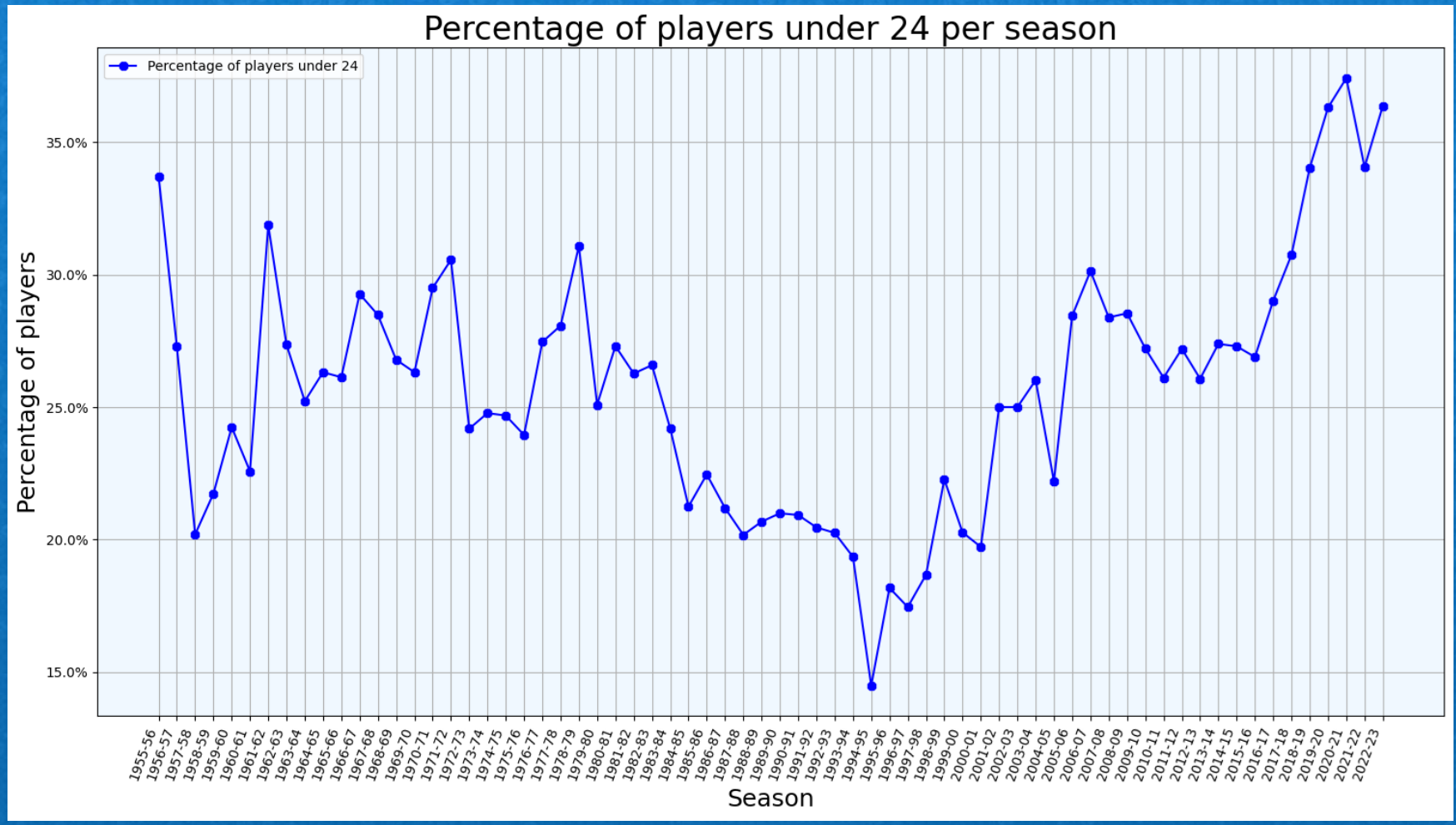
Rk	Player	Pos	Age	Tm	G	GS	MP	FG	FGA	FG%	3P	3PA	3P%	2P	2PA	2P%	eFG%	FT	FTA	FT%	ORB	DRB	TRB	AST	STL	BLK	TOV	PF	PTS
1	Precious Achiuwa	C	23	TOR	55	12	20.7	3.6	7.3	.485	0.5	2.0	.269	3.0	5.4	.564	.521	1.6	2.3	.702	1.8	4.1	6.0	0.9	0.6	0.5	1.1	1.9	9.2
2	Steven Adams	C	29	MEM	42	42	27.0	3.7	6.3	.597	0.0	0.0	.000	3.7	6.2	.599	.597	1.1	3.1	.364	5.1	6.5	11.5	2.3	0.9	1.1	1.9	2.3	8.6
3	Bam Adebayo	C	25	MIA	75	75	34.6	8.0	14.9	.540	0.0	0.2	.083	8.0	14.7	.545	.541	4.3	5.4	.806	2.5	6.7	9.2	3.2	1.2	0.8	2.5	2.8	20.4
4	Ochai Agbaji	SG	22	UTA	59	22	20.5	2.8	6.5	.427	1.4	3.9	.355	1.4	2.7	.532	.532	0.9	1.2	.812	0.7	1.3	2.1	1.1	0.3	0.3	0.7	1.7	7.9
5	Santi Aldama	PF	22	MEM	77	20	21.8	3.2	6.8	.470	1.2	3.5	.353	2.0	3.4	.591	.560	1.4	1.9	.750	1.1	3.7	4.8	1.3	0.6	0.6	0.8	1.9	9.0
6	Nickeil Alexander-Walker	SG	24	TOT	59	3	15.0	2.2	5.0	.444	1.0	2.7	.384	1.2	2.3	.515	.547	0.7	1.0	.667	0.3	1.5	1.7	1.8	0.5	0.4	0.9	1.5	6.2
6	Nickeil Alexander-Walker	SG	24	UTA	36	3	14.7	2.3	4.7	.488	1.0	2.4	.402	1.3	2.3	.578	.591	0.8	1.1	.692	0.2	1.4	1.6	2.1	0.7	0.4	1.3	1.6	6.3
6	Nickeil Alexander-Walker	SG	24	MIN	23	0	15.5	2.1	5.4	.384	1.1	3.1	.361	1.0	2.3	.415	.488	0.6	0.9	.619	0.3	1.5	1.8	1.4	0.3	0.3	0.4	1.3	5.9
7	Grayson Allen	SG	27	MIL	72	70	27.4	3.4	7.7	.440	2.0	5.1	.399	1.4	2.7	.518	.571	1.6	1.8	.905	0.8	2.4	3.3	2.3	0.9	0.2	1.0	1.6	10.4

- Each row is the per-game stats of a player in one season for one team (TOT = all teams combined, for traded players)

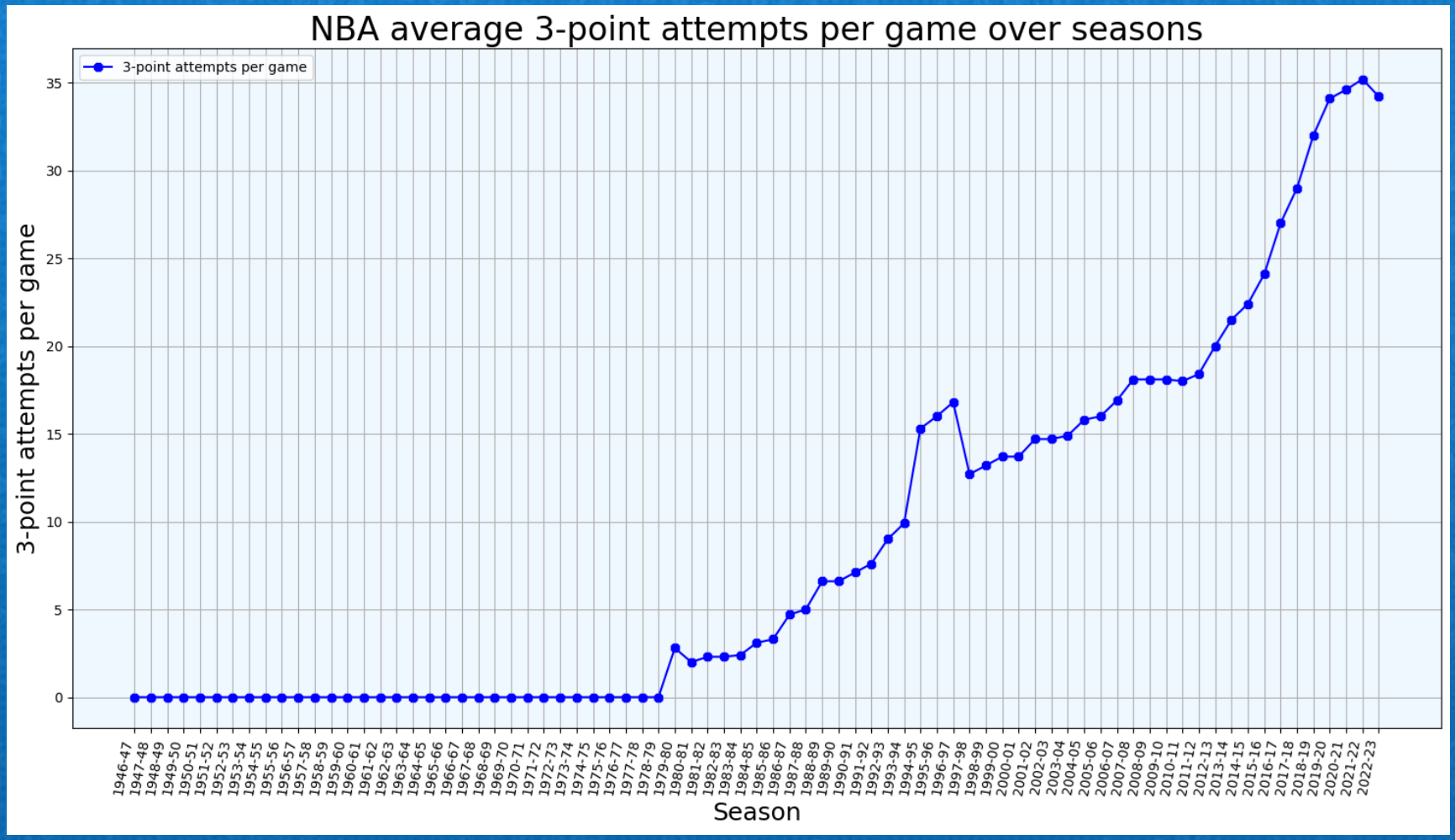
VISUALIZATION OVER TIME



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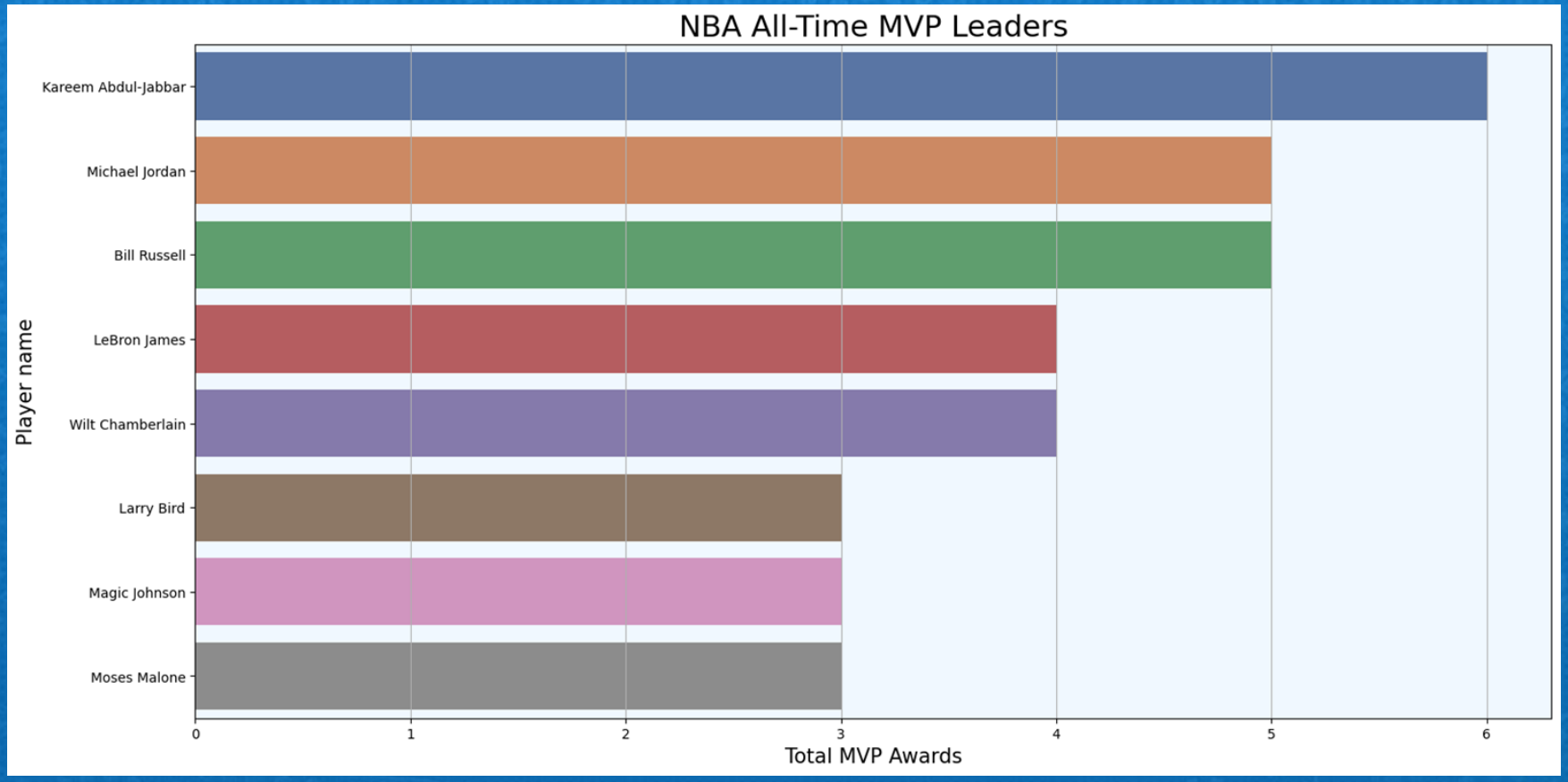
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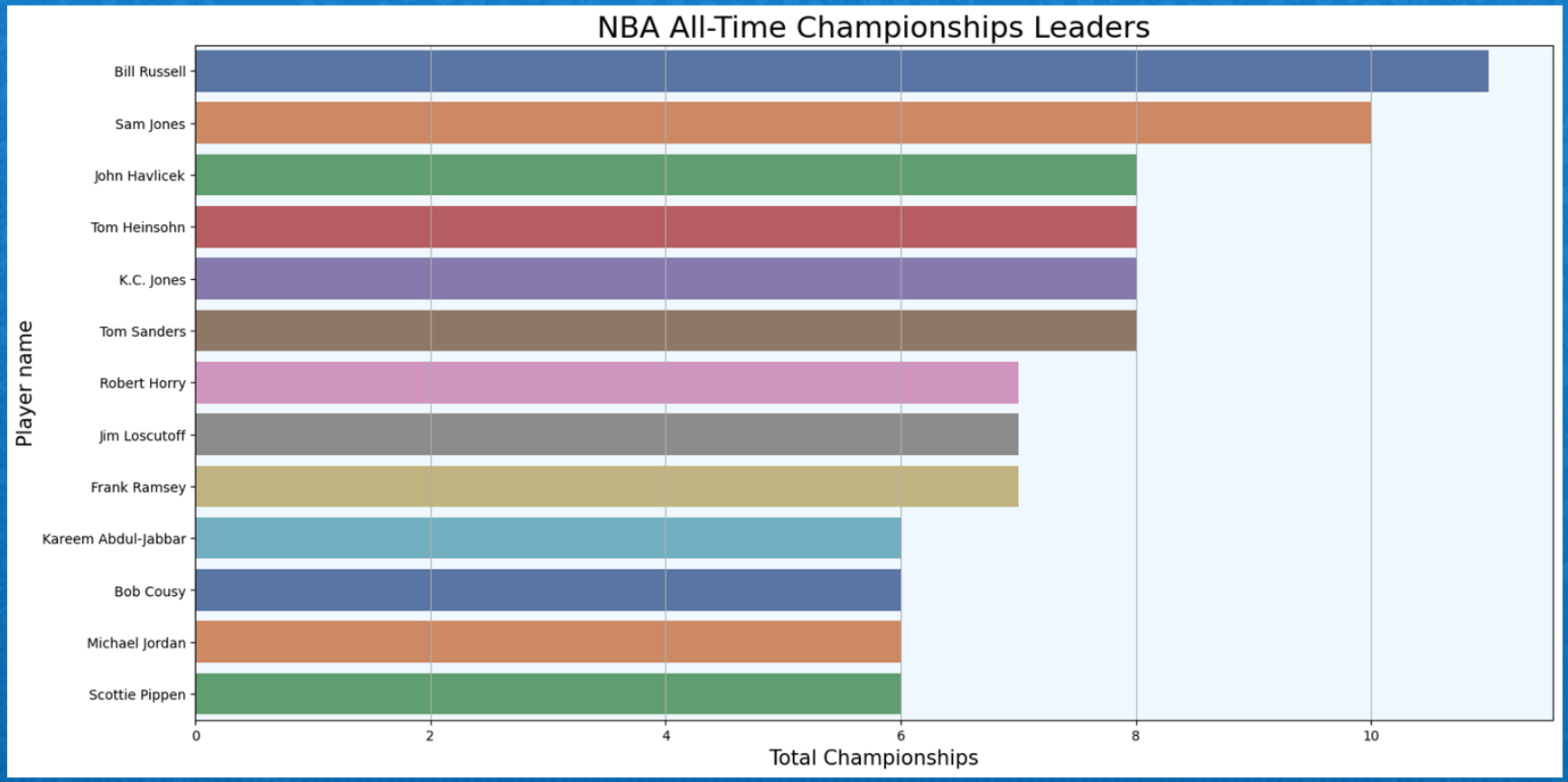
CONCLUSIONS

- After the 1990s with the traditional big men and post-up game focus, the NBA is becoming a young players' league
- The game is evolving, becoming quicker, more possessions (early shot-clock offence) and more 3-point shots (even in transition)
- As a result, young players can make a big difference with their athletic ability and stamina
- The 3-point shot has become a vital skill for players' and teams' success

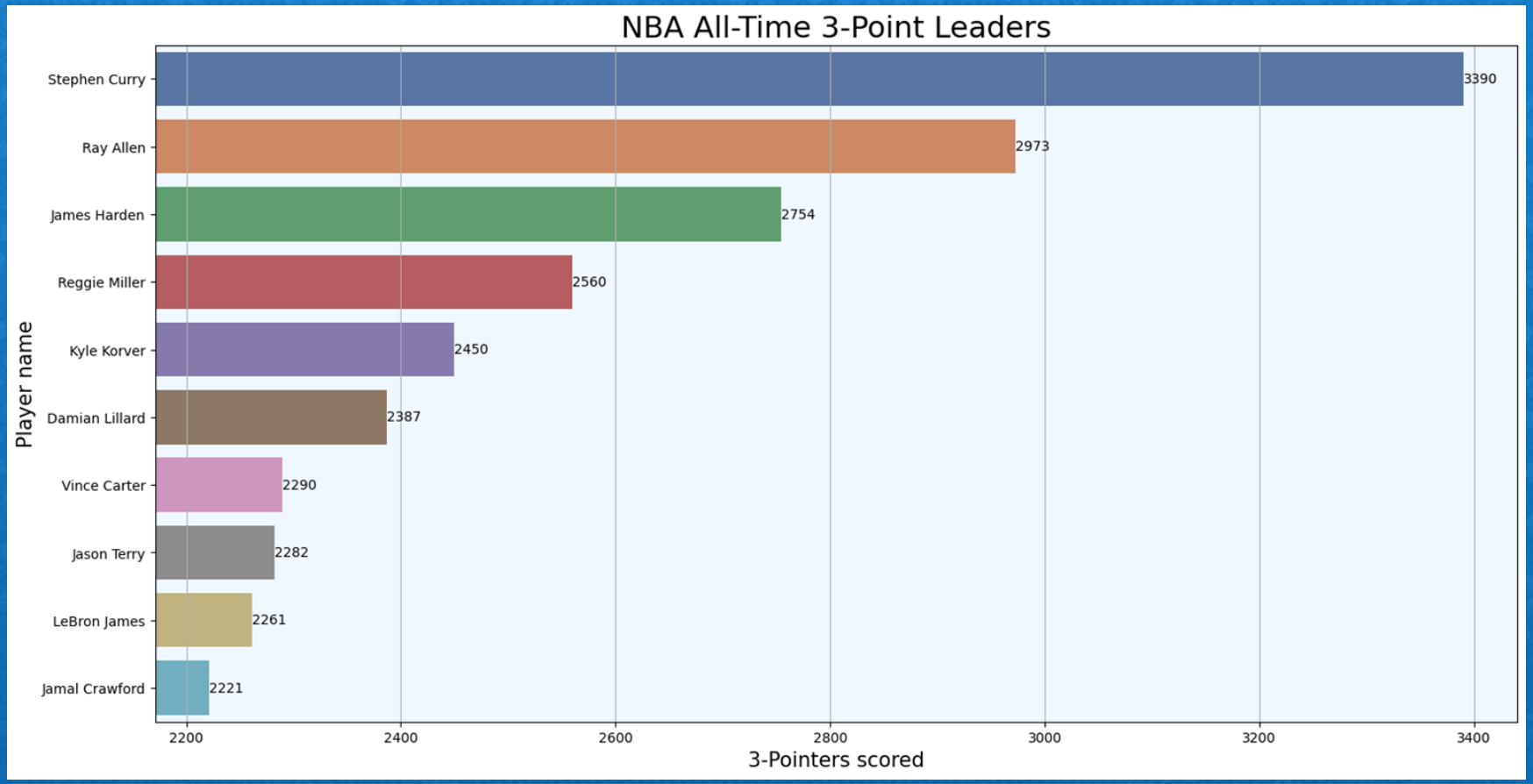
Career Stats Visualization



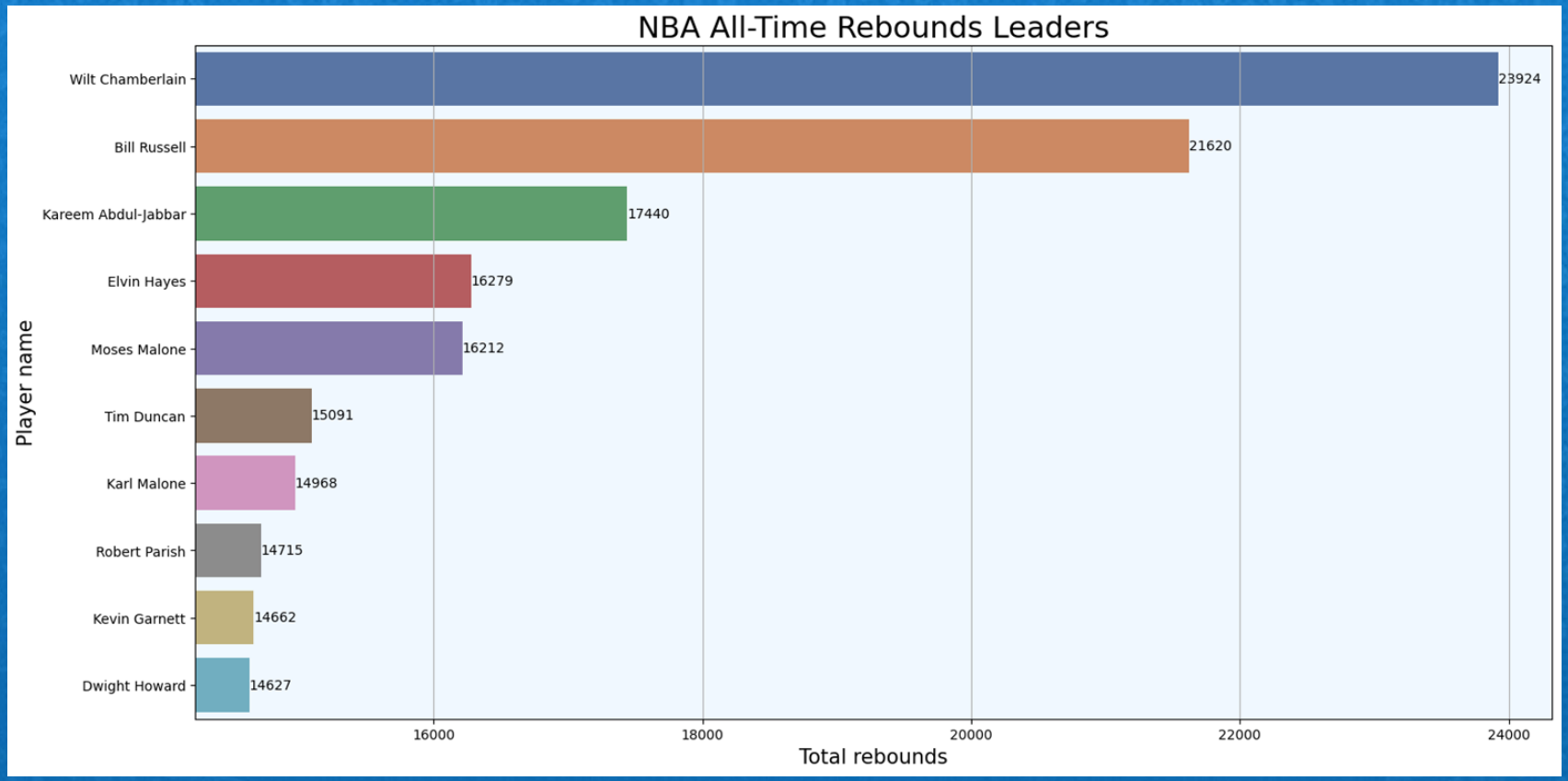
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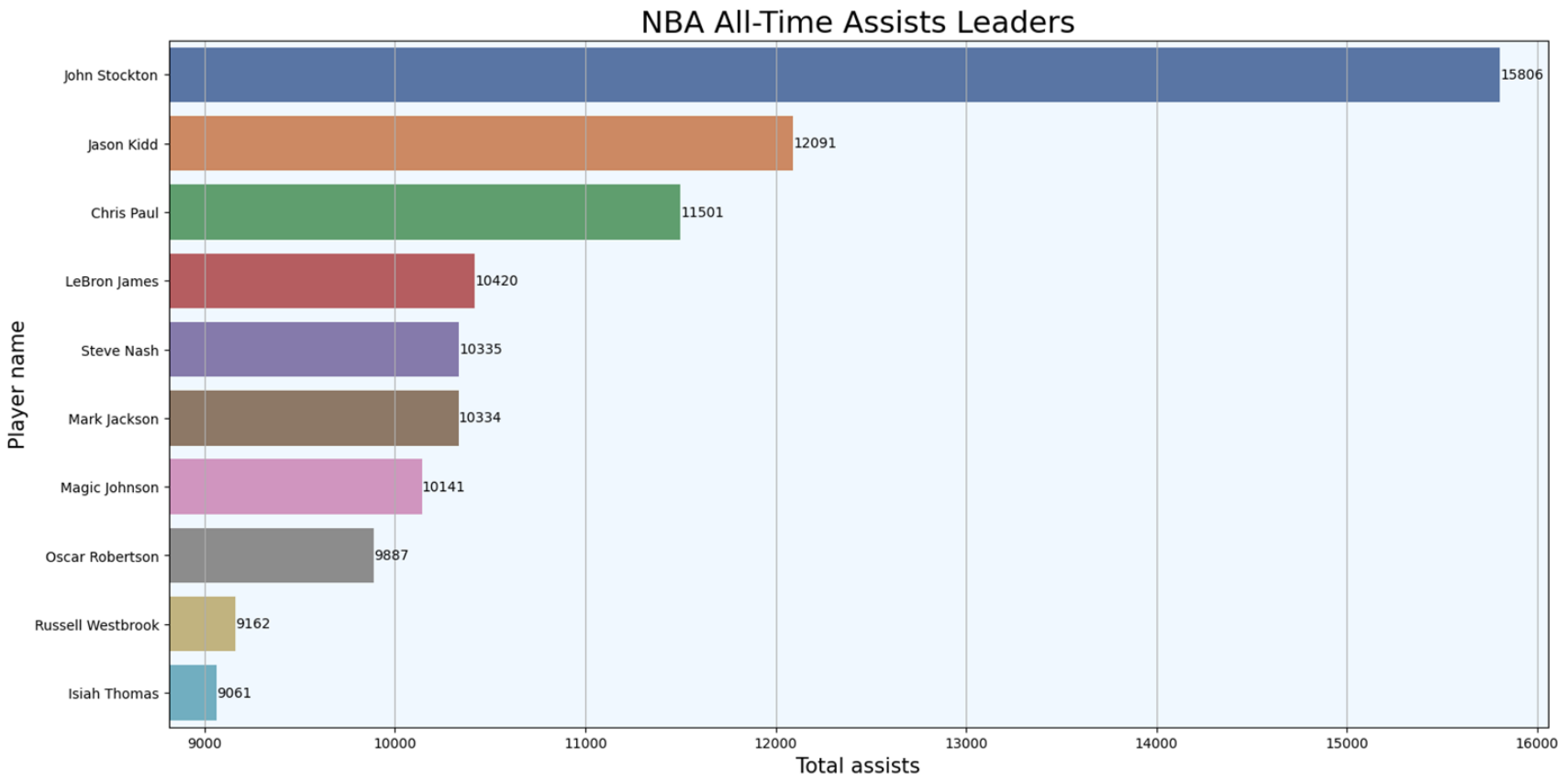
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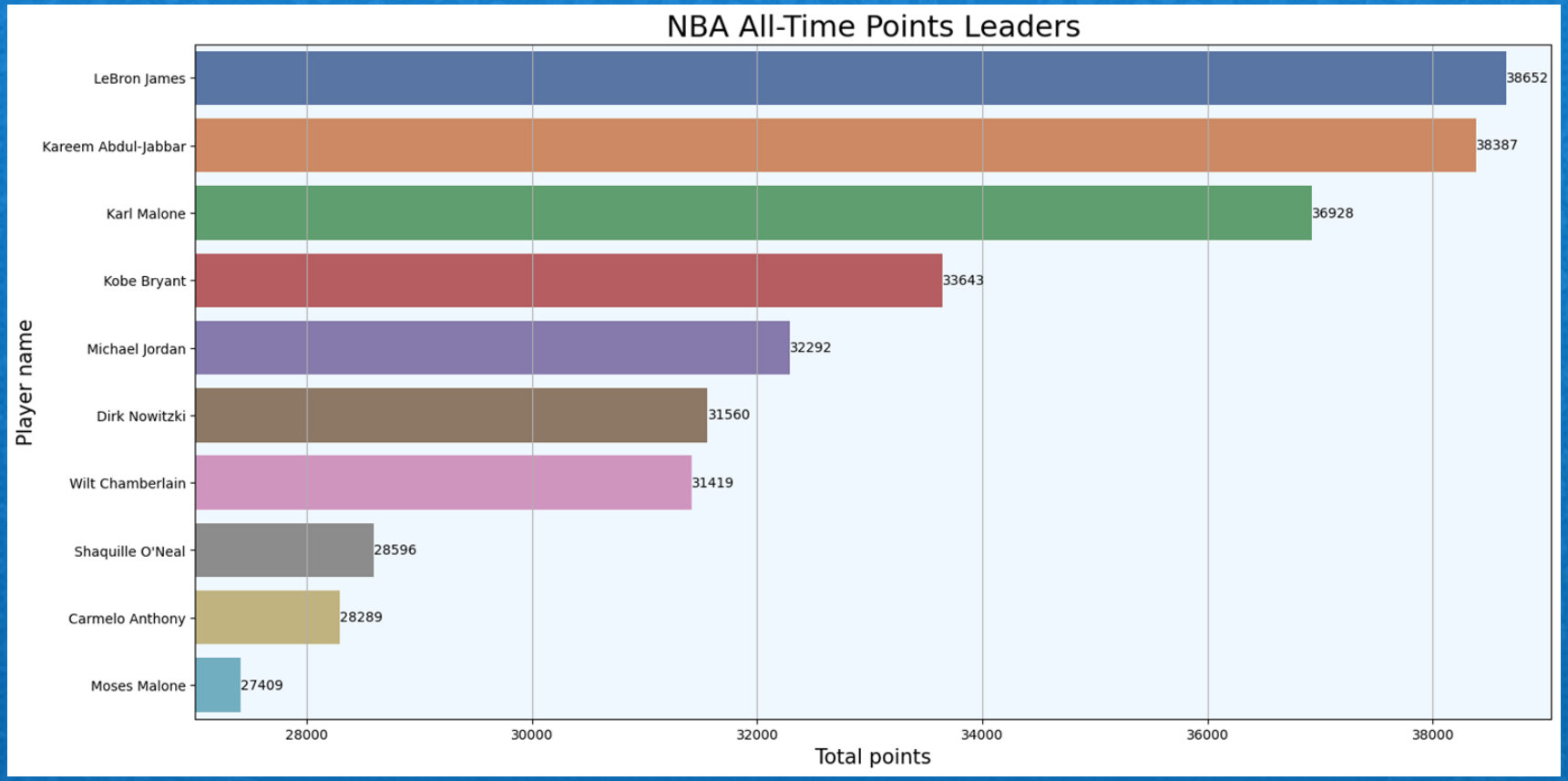
Career Stats Visualization



Career Stats Visualization



Career Stats Visualization



Reading the Career Radar

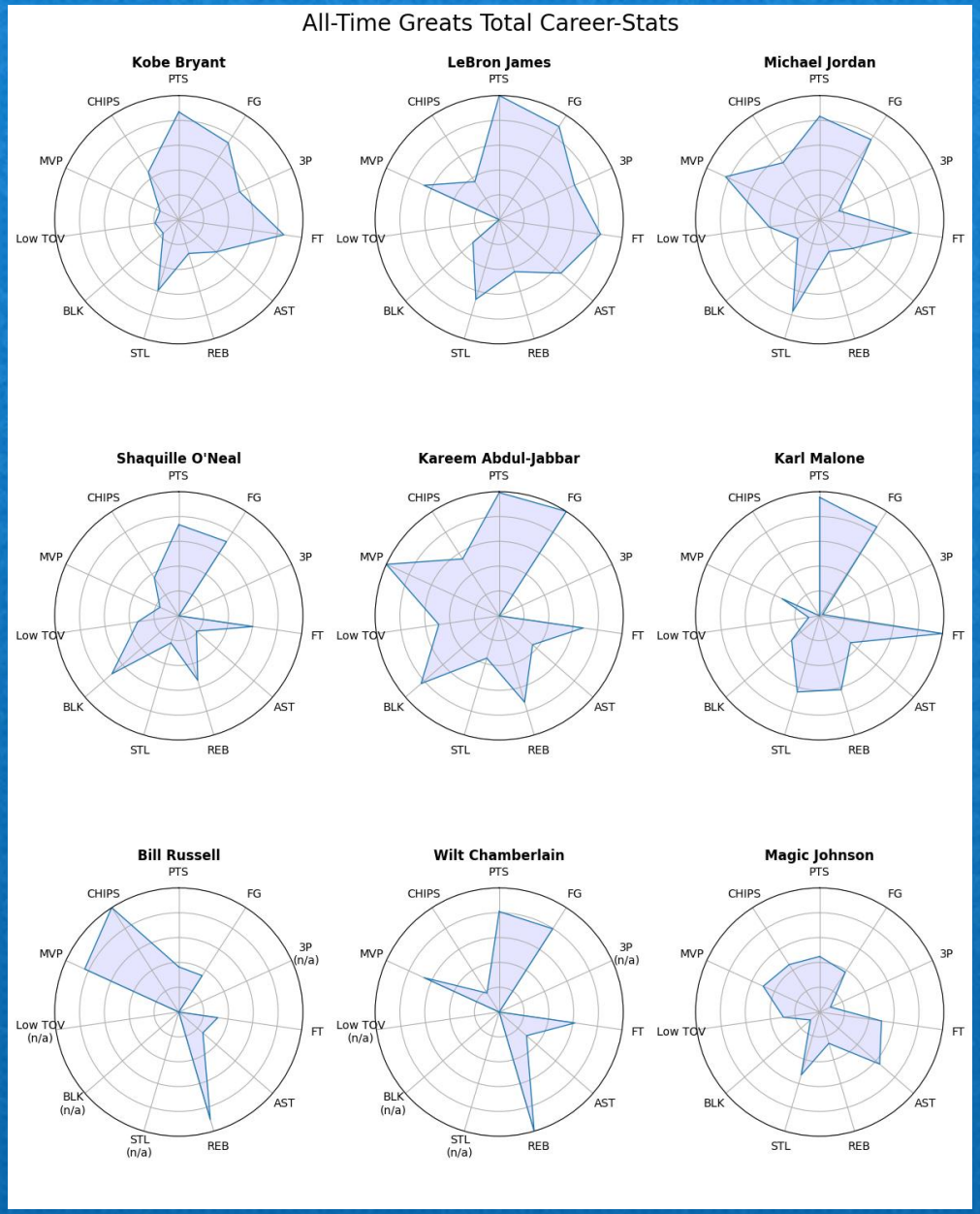
Each axis is a career total, scaled from 0 (centre) to 1 (edge) against all players with 300+ games and 600+ field-goal attempts. n/a means the stat was not recorded during that player's career.

PTS	Total points
FG	Field goals made
3P	3-pointers made (since 1979-80)
FT	Free throws made
AST	Assists
REB	Total rebounds
STL	Steals (since 1973-74)
BLK	Blocks (since 1973-74)
Low TOV	Fewer turnovers (since 1977-78)
MVP	MVP awards
CHIPS	Championships won

Low TOV: further out = fewer turnovers

Turnovers are the only stat where less is better, so the scale is flipped (1 − scaled turnovers). The player with the most career turnovers, LeBron James, sits at the centre. Career totals also grow with longevity.

GOAT Comparison



Reading the Per-Game Radar

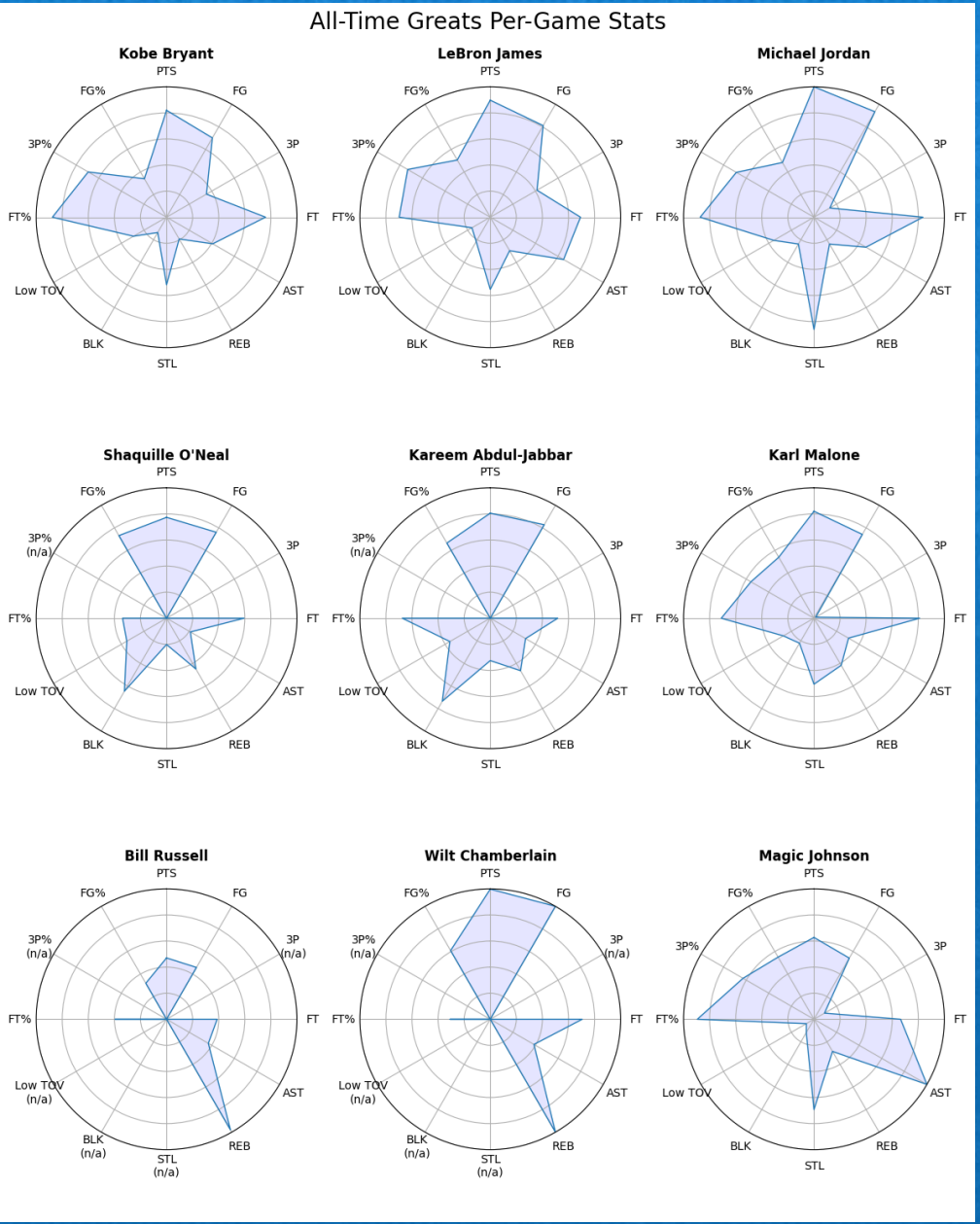
Each axis is a career per-game average or shooting percentage, scaled from 0 (centre) to 1 (edge) against the same players. Steals, blocks, turnovers and 3-pointers are averaged over the games in which they were recorded.

PTS	Points per game
FG	Field goals made per game
3P	3-pointers made per game
FT	Free throws made per game
AST	Assists per game
REB	Rebounds per game
STL	Steals per game
BLK	Blocks per game
Low TOV	Fewer turnovers per game
FT%	Free-throw percentage
3P%	3-point percentage (n/a with 50 or fewer attempts)
FG%	Field-goal percentage

Low TOV: further out = fewer turnovers per game

The axis is flipped in the same way. Players who have the ball in their hands the most, like Magic Johnson and LeBron James, score low here, because more playmaking also means more turnovers.

GOAT Comparison



CONCLUSIONS

- The total-career stats radar-plots are great to see where each great dominated during his career.
- Kareem Abdul-Jabbar and LeBron James have had the greatest NBA careers stat-wise.
- The per-game radar plots show how the greats affected their games, with Michael Jordan the most complete.
- Michael Jordan and LeBron James are head-to-head the greatest players of all time.

REFERENCES

- Complete Jupyter Notebook Analysis
- https://www.basketball-reference.com/

